CHNT31 · Digital Creative Award · Poster

From Pocket to SPARQL

How a Smartphone Scan of an Ogham Stone Learned to Speak Linked Data

Florian Thiery[1,2], **Louise Tharandt**[3]

[1] Leibniz-Zentrum für Archäologie (LEIZA), Mainz, Germany
[2] Research Squirrel Engineers Network
[3] Humboldt-Universität zu Berlin, Berlin, Germany



1 · A scan in the pocket

Citizen-generated 3D models of heritage monuments appear daily on platforms such as Sketchfab. Their metadata are thin, platform-bound and rarely machine-readable, so the models are hard to cite, preserve or link with research data.





The idea: FDOx

FDOx is a practice-driven, Linked-Data-based implementation of the FAIR Digital Object idea (De Smedt et al., 2020) for 3D cultural heritage: one model, one self-describing package (Thiery, 2026). All components are open source and deterministic, so anyone can repeat the workflow for their own models.

Our own perspective, not an official FDO reference implementation.

Explore

FDOx registry · Open-source code · FDOx paper · Ogham stone

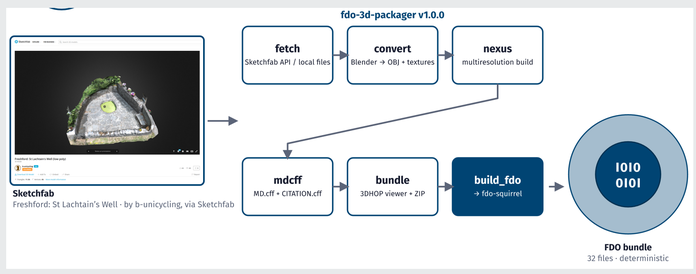
The life of one model: from the pocket to the knowledge graph

1. **Capture** — smartphone scan on Sketchfab
2. **Package** — Python pipeline, open formats, viewer
3. **Describe** — two files become RDF, SHACL-checked
4. **Deposit** — Zenodo with a persistent identifier
5. **Register** — harvest, DCAT catalogue, CIDOC CRM
6. **Query** — SPARQL in the browser; NFDI4Objects feed planned

2 · Package

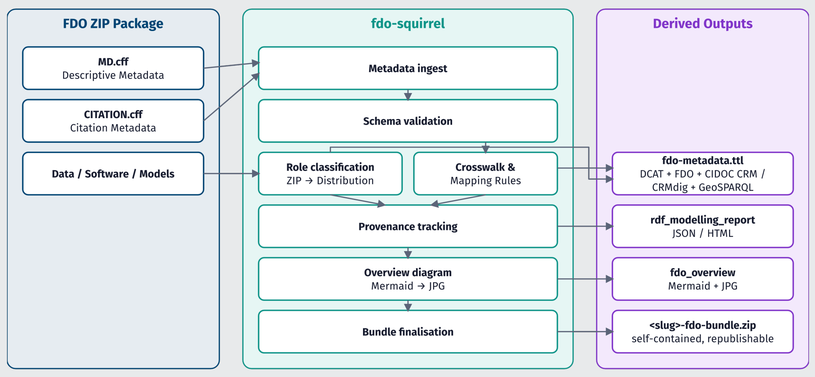
Fetch the model and its metadata from Sketchfab, convert it to open formats, build a web viewer and assemble a self-describing package.

Shown here is fdo-3d-packager v1.0.0 with the Freshford holy well, a model by b-unicycling (CC BY 4.0, via Sketchfab).



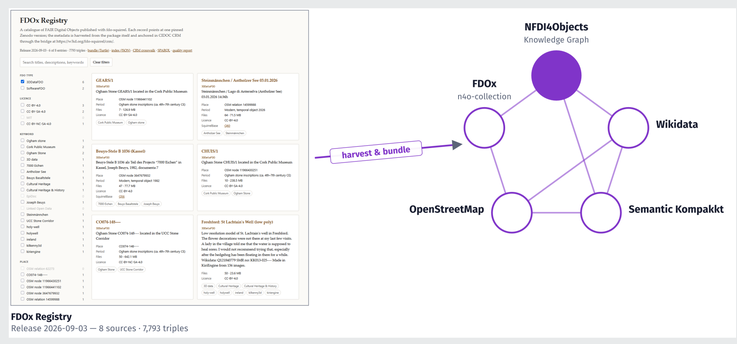
3 · Describe

Two human-editable files, `MD.cff` and `CITATION.cff`, become RDF with DCAT, PROV-O and CIDOC CRM, with provenance for every step; SHACL validates the package.



4–6 · Register and query

Deposited on Zenodo with a persistent identifier, harvested by a registry, anchored in CIDOC CRM and published as a DCAT catalogue that anyone can query with SPARQL in the browser, with no server.



Query 1: which licences?

```
SELECT ?title
(REPLACE(STR(?iri), "^.*/|\\.html$", "") AS ?licence)
WHERE { ?fdo a fdo:3DDataFDO ;
dct:title ?title ; dct:license ?iri .
FILTER(STRSTARTS(STR(?iri), "https://spdx.org/licenses/")) }
ORDER BY ?title
```

Model	Licence
Beuys-Stele B 1036 (Kassel)	CC-BY-4.0
CHUIS/1	CC-BY-SA-4.0
CO074-148---- (CIIC 81)	CC-BY-NC-SA-4.0
Freshford: St Lachtain's Well (low poly)	CC-BY-4.0
GEARS/1	CC-BY-SA-4.0
Steinmännchen / Antholzer See 03.01.2026	CC-BY-4.0

Query 2: where are the 3D models?

```
SELECT ?title ?lat ?lon
WHERE { ?fdo a fdo:3DDataFDO ;
dct:title ?title ;
schema:latitude ?lat ;
schema:longitude ?lon . }
ORDER BY DESC(?lat) ?title
```

Model	Lat	Lon
GEARS/1	51.896151	-8.4942742
CHUIS/1	51.8961	-8.4943
CO074-148---- (CIIC 81)	51.893715	-8.4924208
Beuys-Stele B 1036 (Kassel)	51.3131242	9.4982136
Steinmännchen / Antholzer See	46.888362	12.16988

People

Florian Thiery — **Louise Tharandt**

Behind the work

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References

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- Thiery, F. (2026). FAIR Digital Object Squirrel Implementation (FDOSquirrel). *Squirrel Papers*, 8(2), §4. https://doi.org/10.5281/zenodo.18441544
- fdo-squirrel: https://github.com/FDOx-squirrel/fdo-squirrel
- fdo-3d-packager: https://github.com/FDOx-squirrel/fdo-3d-packager